Cystic Leukoencephalopathy Without Megalencephaly (RNASET2-Deficient Cystic Leukoencephalopathy) — Comprehensive Disease Report

Disease: Cystic Leukoencephalopathy Without Megalencephaly **MONDO ID:** MONDO:0013058 · **OMIM:** #612951 · **Orphanet:** ORPHA:210141 **Causal gene:** *RNASET2* (6q27; HGNC:14015; alias RNASE6PL; NCBI Gene 8635; UniProt O00584) · **Inheritance:** Autosomal recessive **Category:** Mendelian, ultra-rare leukoencephalopathy / type I interferonopathy / lysosomal storage disorder

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Summary

Cystic Leukoencephalopathy Without Megalencephaly is an **ultra-rare, autosomal-recessive infantile leukoencephalopathy** caused by **biallelic loss-of-function variants in *RNASET2*** on chromosome 6q27, which encodes a conserved, glycosylated lysosomal T2-family acid ribonuclease. It was first defined molecularly by Henneke and colleagues in 2009, who recognized that the disorder produces a clinical and neuroradiological picture **indistinguishable from congenital cytomegalovirus (CMV) brain infection** — yet with negative CMV testing — making it a striking **Mendelian mimic** of an acquired congenital infection (PMID: 19525954).

The disease sits at the intersection of three mechanistic classes. It is a **lysosomal storage disorder**: RNase T2 normally degrades ribosomal RNA (rRNA) inside lysosomes, and its loss causes undigested rRNA to accumulate in neuronal lysosomes (demonstrated in zebrafish, PMID: 21199949). It is also a **type I interferonopathy**: the stored lysosomal RNA aberrantly engages endolysosomal RNA-sensing Toll-like receptors (TLR13 in mice, TLR8 inferred in humans), igniting an **IFNAR1-dependent type I interferon response** with microglial

pyroptosis and infiltration of CD8+ T cells and inflammatory monocytes into brain parenchyma (mouse models, PMID: 34764281; PMID: 39853306; PMID: 41453865). This places it in clinical and pathological overlap with **Aicardi–Goutières syndrome (AGS)** (PMID: 27091087).

Clinically, the disorder presents in infancy with a **largely static, severe encephalopathy** — profound psychomotor impairment, spasticity, epilepsy, and sometimes microcephaly, hearing loss, or dystonia. The neuroradiological hallmarks are **bilateral anterior temporal subcortical cysts, multifocal lobar white-matter lesions with sparing of central white matter, and intracranial calcification**. Diagnosis is molecular, since imaging alone cannot separate it from congenital CMV or AGS. There is **no disease-specific therapy**; management is supportive, though the interferonopathy mechanism nominates **JAK1/2 inhibition** (baricitinib, ruxolitinib) as a biologically rational but as-yet-untrialed candidate. Fewer than a few dozen families have been reported worldwide.

Key Findings

1. *RNASET2* biallelic loss-of-function is the sole known cause and produces a CMV-mimic phenotype

The founding study (Henneke et al., 2009) mapped and identified **homozygous and compound heterozygous loss-of-function mutations in *RNASET2*** at chromosome 6q27 as the cause of an autosomal-recessive cystic leukoencephalopathy whose clinical and neuroradiological phenotype is *indistinguishable from congenital CMV brain infection*. The verbatim conclusion: *"loss-of-function mutations in the gene encoding the RNASET2 glycoprotein lead to cystic leukoencephalopathy, an autosomal recessive disorder with an indistinguishable clinical and neuroradiological phenotype"* (PMID: 19525954). The disorder carries **OMIM #612951** and **MONDO:0013058**. Multiple subsequent families have confirmed the gene–disease relationship, including Tonduti et al. 2016 (PMID: 27091087) and Sun et al. 2018 (PMID: 29336640). *(Evidence: human clinical/genetic.)*

2. The disease is a lysosomal storage disorder in which rRNA is the stored material

Haud et al. (2011) generated *rnaset2* mutant zebrafish and showed that **RNase T2 localizes within lysosomes and that its loss causes accumulation of undigested rRNA inside neuronal lysosomes**: *"loss of rnaset2 in mutant zebrafish results in accumulation of undigested rRNA within lysosomes within neurons of the brain."* High-field MR microimaging revealed white-matter lesions comparable to those in RNASET2-deficient infants, together with amyloid precursor protein accumulation and astrogliosis at sites of neurodegeneration. The authors concluded that *"familial cystic leukoencephalopathy is a lysosomal storage disorder in which rRNA is the best candidate for the noxious storage material"* (PMID: 21199949). This finding establishes both the **subcellular site (lysosome, GO:0005764)** and the **initiating molecular lesion**. *(Evidence: model organism — zebrafish.)*

3. RNASET2 deficiency is a type I interferonopathy driven by endolysosomal RNA-sensing TLR activation

Two 2025 companion studies converge on **TLR13 as the driver of RNase T2-deficient autoinflammation in mice**. Gomez-Diaz et al. showed that *Rnaset2-/-* mice develop interferon-dependent neuroinflammation, impaired hematopoiesis, and splenomegaly, and that *"the inflammatory phenotype found in Rnaset2-/- mice is completely reversed in the absence of TLR13, suggesting aberrant accumulation of an RNA ligand for this receptor"* (PMID: 39853306). Crucially, the phenotype persists in germ-free mice, indicating an **endogenous rRNA-derived ligand**. Sato et al. showed that *"lysosomal RNA stress, caused by the lack of RNase T2, induces macrophage accumulation in multiple organs such as the spleen and liver through TLR13 activation by microbiota-derived ribosomal RNAs"* (PMID: 39853307). Because humans lack functional TLR13, **TLR8** is the inferred orthologous endolysosomal RNA sensor in patients. *(Evidence: model organism — mouse.)*

4. Microglial pyroptosis and ISG upregulation precede T-cell infiltration and brain atrophy

Wendland et al. (2025) charted the temporal cascade in *Rnaset2-/-* mice. Interferon-stimulated genes (IRF9, RIG-I) are sustainedly upregulated across 3–28 weeks; chemokines *Ccl2*, *Ccl5*, *Cxcl10* peak early; and **pyroptosis markers ASC, CASP1, and GSDMD are significantly increased at 3–6 weeks** (declining thereafter) while apoptotic markers (Bax, CASP3/8, PARP) remain unchanged. ASC co-localizes with the microglial marker IBA-1, and *Cd3e*/*Tnf* peak later

(~17 weeks). The authors conclude that *"pyroptosis is an early, disease-associated event restricted to microglia that likely contributes to establishing a proinflammatory milieu prior to T cell infiltration and brain atrophy"* (PMID: 41453865). This positions **microglial pyroptosis as an early, upstream driver** of neurodegeneration and nominates inflammasome/pyroptosis inhibition as a therapeutic node. *(Evidence: model organism — mouse.)*

5. IFNAR1 dependence proves type I interferon causality

Kettwig et al. (2021) generated CRISPR/Cas9 *Rnaset2*-/- mice and demonstrated that neuroinflammation is **IFNAR1-dependent**: *"Rnaset2-/- mice demonstrate upregulation of interferon-stimulated genes and concurrent IFNAR1-dependent neuroinflammation, with infiltration of CD8+ effector memory T cells and inflammatory monocytes into the grey and white matter."* Single-nuclei RNA sequencing revealed *"homeostatic dysfunctions in glial cells and neurons,"* and the mice showed hippocampal-accentuated brain atrophy with cognitive impairment (PMID: 34764281). Genetic removal of the type I interferon receptor abrogating the phenotype provides the **causal proof** that type I IFN signaling — not merely storage — drives the neuropathology. *(Evidence: model organism — mouse.)*

6. Characteristic clinical and neuroradiological phenotype

Across reported families, the disorder is an **infantile-onset, largely static encephalopathy** with severe psychomotor impairment/developmental delay (the cardinal feature), spasticity, epilepsy/seizures, occasional neurological regression, and **microcephaly — explicitly NOT megalencephaly**. The signature MRI triad is **bilateral anterior temporal subcortical cysts, multifocal lobar white-matter lesions with sparing of central white matter, and intracranial calcification**: Tonduti et al. describe *"bilateral anterior temporal subcortical cysts and multifocal lobar white matter lesions with sparing of central white matter structures"* (PMID: 27091087), and Kameli et al. report *"white matter involvement, calcification and anterior temporal cysts"* (PMID: 31349848). CMV PCR is negative and metabolic screening is normal, distinguishing the disorder from its acquired mimic. Suggested HPO terms: intellectual disability (HP:0001249), global developmental delay (HP:0001263), spasticity (HP:0001257), seizure (HP:0001250), microcephaly (HP:0000252), cerebral white matter atrophy/ leukoencephalopathy (HP:0002352), intracranial calcification (HP:0002514), sensorineural hearing impairment (HP:0000407), dystonia (HP:0001332). *(Evidence: human clinical.)*

7. Gene, protein, and representative pathogenic variants

RNASET2 encodes a **256-amino-acid acidic ribonuclease** of the conserved T2 family — a glycoprotein located on chromosome 6q27 (PMID: 31349848: *"RNASET2 as a subtype of RNASEs is a 256 amino acid protein, encoded by RNASET2 gene located on chromosome six"*). Inheritance is autosomal recessive with biallelic loss-of-function variants. Reported pathogenic variants include the nonsense variants **c.233C>A p.(Ser78Ter)** (PMID: 31349848) and **c.128G>A p.(Trp43Ter)** (PMID: 29336640), plus a **430-kb 6q27 microdeletion encompassing *RNASET2*** and the compound-heterozygous/homozygous LoF alleles of the founding study (PMID: 19525954). The protein (UniProt **O00584**; alias RNASE6PL) carries two catalytic active-site histidines (CAS I/CAS II). *(Evidence: human clinical/genetic.)*

8. RNASET2 is a conserved alarmin and tumor suppressor beyond its housekeeping role

Beyond lysosomal rRNA turnover, human RNASET2 is a **secreted "alarmin" and oncosuppressor** that recruits and activates monocyte/macrophage-lineage innate immune cells. Rosini et al. describe that *"the human RNASET2 protein (hRNASET2) has been reported as an extracellular tumor suppressor protein, endowed with the ability to act as an 'alarmin' signalling molecule"* (PMID: 32450138), and Lualdi et al. document its role in *"inducing a sustained recruitment of immune-competent cells belonging to the monocyte/macrophage lineage within a growing tumor mass"* (PMID: 25797262). This dual identity — housekeeping lysosomal RNase and innate-immune signaling molecule — helps explain why its loss produces both storage pathology and inflammation. *(Evidence: in vitro / cell biology.)*

9. Phenotypic spectrum is variable; anterior temporal cysts are not obligate

The RNASET2-deficiency spectrum is broader than the classic triad. Sun et al. explicitly note that their patient *"did not show anterior temporal lobe subcortical cysts, hearing loss, dystonia or extra-neurological features"* (PMID: 29336640) — implying that **sensorineural hearing loss and dystonia occur in other patients** and that the hallmark anterior temporal cysts are **variably present**. Core features (delayed psychomotor development, intellectual disability, seizures) are consistent, but severity and the full imaging picture vary, indicating **variable expressivity**. *(Evidence: human clinical.)*

10. Ultra-rare, consanguinity-enriched, reported across diverse populations

As of 2018, *"Only eight families with RNASET2 mutation have been previously reported"* (PMID: 29336640), with additional families reported since (e.g., Tonduti 2016, 5 patients; Kameli 2019). Cases span European, East Asian/Chinese, and Iranian populations. Consanguinity contributes homozygous LoF alleles and structural microdeletions. Orphanet lists prevalence as unknown/ <1 per 1,000,000. *(Evidence: human epidemiological.)*

11. Deep evolutionary conservation and three engineered animal models

The T2/Rh ribonuclease family is ancient and ubiquitous: *"T2-family acidic endoribonucleases are represented in all genomes"* (PMID: 21199949). Verified orthologs include human *RNASET2* (GeneID 8635; Taxon 9606), mouse *Rnaset2a*/*Rnaset2b* (GeneIDs 100037283/68195; Taxon 10090), rat *Rnaset2* (GeneID 292306; Taxon 10116), and zebrafish *rnaset2* (GeneID 791890; Taxon 7955). Three engineered models exist — **zebrafish** (lysosomal rRNA storage + white-matter lesions), **mouse** (IFNAR1-dependent neuroinflammation, cystic/white-matter lesions, atrophy), and **rat** (hippocampal neuroinflammation and memory deficits but no cystic lesions; PMID: 29752287). No naturally occurring RNASET2 disease is documented in OMIA. *(Evidence: model organism / comparative.)*

Mechanistic Model / Interpretation

Ordered causal chain (initiating lesion → clinical manifestation)

1. **Biallelic loss-of-function variants in *RNASET2*** (6q27; nonsense, deletion, or microdeletion) **result in** absent/nonfunctional lysosomal T2-family acid ribonuclease. *(Demonstrated — human genetics,* P *19525954 .)*
2. Loss of RNase T2 catalytic activity **leads to** failure of lysosomal ribosomal RNA degradation, so **undigested rRNA accumulates within neuronal lysosomes** — a lysosomal storage state. *(Demonstrated — zebrafish,* P *21199949 .)*
3. Accumulated lysosomal rRNA **acts as a ligand for endolysosomal RNA-sensing Toll-like receptors — TLR13 in mice** (genetically proven), **TLR8 inferred in humans**.

(Demonstrated in mouse; human step inferred —

P 39853306

P 39853307

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4. TLR activation **drives type I interferon production and signaling**, which is **IFNAR1-dependent**: removing IFNAR1 abrogates neuroinflammation. *(Demonstrated — mouse,*

   P 34764281

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5. Type I IFN signaling **results in** sustained interferon-stimulated gene (ISG) upregulation and **early microglial pyroptosis** (ASC/CASP1/GSDMD), establishing a proinflammatory milieu. *(Demonstrated — mouse,*

   P 41453865

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6. The proinflammatory milieu **leads to** infiltration of **CD8+ effector-memory T cells and inflammatory monocytes** into grey and white matter (branch: also splenomegaly, impaired hematopoiesis, emergency myelopoiesis systemically). *(Demonstrated — mouse,*

   P 34764281

   P 39853307

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7. Glial/neuronal homeostatic dysfunction and neuroinflammation **cause** white-matter injury, anterior temporal subcortical cysts, intracranial calcification, and cerebral (hippocampal-accentuated) atrophy. *(Demonstrated across models + human imaging.)*

8. These structural lesions **produce** the clinical phenotype: severe, largely static infantile encephalopathy with psychomotor impairment, spasticity, epilepsy, ± microcephaly/hearing loss/dystonia. *(Demonstrated — human clinical,*

   P 19525954

   P 27091087

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RNASET2 biallelic LoF (6q27)
           │  loss of lysosomal acid RNase
           ▼
Undigested rRNA storage in neuronal lysosomes  ── lysosomal storage disorder
           │  endogenous RNA ligand
           ▼
Endolysosomal RNA-sensing TLR (TLR13 mouse / TLR8 human inferred)
           │
           ▼
Type I interferon production ──► IFNAR1-dependent signaling ── type I interferonopathy
           │
           ├──► ISG upregulation (IRF9, RIG-I)
           ├──► EARLY microglial pyroptosis (ASC/CASP1/GSDMD)  [3–6 wk peak]
           │
           ▼
CD8+ T-cell + inflammatory monocyte infiltration [~17 wk]
           │                                    └─(systemic: splenomegaly, myelopoiesis)
           ▼
White-matter injury · anterior temporal cysts · calcification · atrophy
           │
           ▼
Severe static infantile encephalopathy (CMV/AGS mimic)
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Upstream vs downstream: The lysosomal storage defect (steps 1–2) is upstream and cell-intrinsic; the TLR→IFN axis (steps 3–5) is the amplifying inflammatory core; T-cell/monocyte infiltration and tissue injury (steps 6–7) are downstream effectors. The IFNAR1-knockout rescue (step 4) and TLR13-knockout rescue (step 3) identify two genetically validated intervention points.

Cell types involved (CL terms): neurons (CL:0000540), oligodendrocytes/myelin (CL:0000128), astrocytes/astrogliosis (CL:0000127), microglia (CL:0000129), infiltrating CD8+ T cells, inflammatory monocytes/macrophages. **Biological processes (GO terms):** lysosomal RNA catabolism (rRNA degradation), Toll-like receptor signaling pathway (GO:0002224), type I interferon-mediated signaling pathway (GO:0060337), pyroptosis (GO:0070269) / inflammasome activation, neuroinflammatory response (GO:0150076). **Subcellular compartment (GO CC):** lysosome (GO:0005764) — the site of the primary storage lesion.

Comparison of the three axes of pathogenesis

Mechanistic class	Evidence	Key node	Therapeutic implication
Lysosomal storage disorder	Zebrafish rRNA accumulation (PMID: 21199949)	Failed rRNA degradation	Substrate reduction / enzyme replacement (theoretical)
Type I interferonopathy	IFNAR1-dependence (PMID: 34764281); TLR13 rescue (PMID: 39853306)	TLR→IFN axis	JAK1/2 inhibition; TLR blockade
Innate-immune / pyroptotic	Microglial pyroptosis (PMID: 41453865)	Inflammasome (ASC/CASP1/ GSDMD)	Inflammasome/ pyroptosis inhibitors

Anatomical, Temporal, and Population Summary

Anatomy (Section 7 of template): Primary organ — brain (UBERON:0000955), nervous system. Sites — bilateral cerebral/lobar white matter (UBERON:0002316) with central white-matter sparing, anterior temporal lobe subcortical cysts (temporal lobe UBERON:0001871), basal ganglia/intracranial calcification, and cerebral atrophy including hippocampus (UBERON:0002421). Lateralization — **bilateral/symmetric**. Kettwig et al. confirm *"cystic brain lesions, multifocal white matter alterations, cerebral atrophy"* (PMID: 34764281).

Temporal development (Section 8): Onset is **congenital/infantile**; course is **largely static** (non- or slowly progressive) rather than relentlessly degenerative, though regression is described in some patients. Mouse data reveal a defined temporal order — ISGs and chemokines early, microglial pyroptosis peaking at 3–6 weeks, T-cell infiltration ~17 weeks — suggesting an **early critical window** for anti-inflammatory intervention before irreversible atrophy.

Inheritance & population (Section 9): Autosomal recessive; ultra-rare (<1/1,000,000). Consanguinity-enriched; founder-type homozygous LoF and microdeletions occur. Reported in European, Chinese/East Asian, and Iranian families. No strong sex bias reported. Penetrance appears high for biallelic LoF; expressivity is variable (see Finding 9). No genetic anticipation, mosaicism, or repeat-expansion mechanism is implicated.

Diagnostics (Section 10): Diagnosis is **molecular** (single-gene *RNASET2* testing, leukodystrophy gene panels, WES, or WGS; chromosomal microarray detects the 6q27 microdeletion), because MRI cannot distinguish the disorder from congenital CMV or AGS. Supportive workup: **negative CMV PCR**, normal metabolic screening, characteristic MRI triad. Differential diagnosis — congenital CMV infection, Aicardi–Goutières syndrome, and other cystic leukoencephalopathies (e.g., megalencephalic leukoencephalopathy, which by contrast features macrocephaly). A blood interferon signature (ISG score) would be expected to support the interferonopathy classification, though it is not yet validated as a routine test in this disorder.

Prognosis (Section 11): Severe neurodevelopmental disability; course largely static with lifelong dependency. Morbidity is high (spasticity, epilepsy, cognitive impairment). Formal survival statistics are not established given rarity.

Treatment (Section 12): No disease-specific therapy. Care is symptomatic/supportive — antiepileptics (NCIT anticonvulsant agents), spasticity management (e.g., baclofen), physiotherapy/occupational/speech therapy, developmental support. **JAK1/2 inhibitors** (baricitinib, ruxolitinib, tofacitinib) are a mechanistically rational, untrialed candidate given the interferonopathy classification; a scoping review of type I interferonopathies found that *"JAK inhibitors improved clinical and analytical parameters and decreased flare numbers, plasma inflammatory markers, and expression of IFN-stimulated genes"* (PMID: 33856640). Preclinical work additionally nominates inflammasome/pyroptosis inhibition (PMID: 41453865).

Prevention (Section 13): No primary prevention exists. Relevant measures are **genetic counseling**, carrier testing in consanguineous families, cascade testing, and prenatal/preimplantation genetic testing where a familial variant is known.

Other species / models (Sections 14–15): No naturally occurring animal disease documented in OMIA. Three engineered models — **zebrafish** (faithful lysosomal storage + white-matter lesions), **mouse** (best recapitulation: interferonopathy, cystic/white-matter pathology, atrophy, cognitive deficits), and **rat** (hippocampal neuroinflammation and memory deficits, but lacks cystic lesions). A 2024 review documents zebrafish as a leukodystrophy model (PMID: 38239149).

Evidence Base

PMID	Title (abbrev.)	Evidence type	Supports
19525954	*RNASET2-deficient cystic leukoencephalopathy resembles congenital CMV*	Human genetics/clinical	Gene–disease causality; CMV mimic (F001, F005, F006)
21199949	*rnaset2 mutant zebrafish model familial cystic leukoencephalopathy...*	Model organism (zebrafish)	Lysosomal rRNA storage mechanism; conservation (F002, F011, F013)
27091087	*Clinical, radiological and pathological overlap ... and AGS*	Human clinical	MRI triad; AGS overlap (F005, F008)
29336640	*Novel RNASET2 pathogenic variants in an East Asian child*	Human genetics/clinical	Variants; rarity; phenotypic variability (F006, F009, F010)
31349848	Kameli et al., RNASET2 case	Human clinical/ genetic	Protein size/location; variant; imaging (F005, F006)
34764281	Kettwig et al., *Rnaset2-/- mice*	Model organism (mouse)	IFNAR1-dependence; cellular infiltrate (F011, F012, F014)
39853306	Gomez-Diaz et al., TLR13	Model organism (mouse)	TLR13 as driver; interferonopathy (F003, F014)
39853307	Sato et al., lysosomal RNA stress	Model organism (mouse)	TLR13/rRNA link; myelopoiesis (F003)
41453865	Wendland et al., microglial pyroptosis	Model organism (mouse)	Early pyroptosis cascade (F004)
29752287	Sinkevicius et al., RNaseT2 KO rat	Model organism (rat)	Rat model characteristics (F011)
25797262	Lualdi et al., RNASET2 alarmin	In vitro	Alarmin/immune recruitment (F007)
32450138	Rosini et al., hRNASET2 tumor suppressor	In vitro	Alarmin/tumor-suppressor identity (F007)

PMID	Title (abbrev.)	Evidence type	Supports
33856640	Gómez-Arias et al., JAK inhibitors in interferonopathies	Human clinical (scoping review)	Rational therapy (F008)
38239149	Review — zebrafish leukodystrophy models	Review	Model relevance (F014)

How the evidence coheres: Human genetics (P 19525954) established causality; zebrafish (P 21199949) defined the storage mechanism; and a series of rodent models (P 34764281 / P 39853306 07, 41453865) built the inflammatory arc, culminating in two genetically validated intervention points (TLR13, IFNAR1). No study contradicts the consolidated model; the primary uncertainty is translational (the human TLR8-vs-mouse-TLR13 inference).

Limitations and Knowledge Gaps

- **Human TLR identity unproven.** The TLR13 dependence is established only in mice; humans lack functional TLR13, so the orthologous sensor (**TLR8** is the leading candidate) has not been experimentally confirmed in patient cells. This is the single most important translational gap.
- **No human trial data for any disease-modifying therapy.** JAK inhibition and inflammasome/pyroptosis inhibition are mechanistically rational but entirely untrialed in this disease; efficacy, blood–brain-barrier penetration, and timing (early critical window) are unknown.

- **Small evidence base for epidemiology and natural history.** With only a few dozen reported families, prevalence, penetrance, expressivity, sex ratio, and survival statistics are imprecise. No formal registry or natural-history cohort exists.
- **Genotype–phenotype correlations are undefined.** It is unclear why some patients lack anterior temporal cysts or develop hearing loss/dystonia while others do not; modifier genes and environmental modifiers are unexplored.
- **Static-vs-progressive course.** The relative contributions of a fixed developmental lesion versus ongoing inflammation to the "static" clinical picture are not fully resolved, which matters for whether late anti-inflammatory therapy could help.
- **Human neuropathology is limited.** Most cellular/mechanistic detail derives from animal models; direct confirmation of microglial pyroptosis and the ISG signature in human brain tissue is sparse.
- **No epigenetic, proteomic, or metabolomic patient datasets** were identified for this disease; molecular profiling remains model-based.

Proposed Follow-up Experiments / Actions

1. **Confirm the human sensor.** Test whether patient-derived macrophages/microglia (iPSC-derived) or *RNASET2*-knockout human cells show TLR8-dependent type I IFN induction; use TLR8 antagonists and CRISPR knockout to establish the human ortholog of the mouse TLR13 axis.
2. **Preclinical JAK-inhibitor trial in *Rnaset2*-/- mice.** Test whether baricitinib/ruxolitinib started within the early critical window (before the 3–6-week pyroptosis peak) prevents ISG upregulation, T-cell infiltration, and hippocampal atrophy, with dose–timing arms to define the intervention window.
3. **Inflammasome/pyroptosis blockade.** Evaluate GSDMD or CASP1/NLRP3 inhibitors (or genetic *Gsdmd* deletion) in *Rnaset2*-/- mice to test whether interrupting microglial pyroptosis is sufficient to blunt downstream neuroinflammation.
4. **Interferon signature as a biomarker.** Measure a blood ISG score (IFI27, IFI44L, IFIT1, ISG15, etc.) in RNASET2-deficient patients to confirm the human interferonopathy signature and provide a pharmacodynamic readout for future trials.

5. **Establish an international registry / natural-history study** to capture prevalence, genotype–phenotype correlations (including the variable temporal cysts, hearing loss, dystonia), progression, and survival.
6. **Substrate-reduction and enzyme-replacement feasibility.** Explore whether reducing lysosomal rRNA load (e.g., modulating ribophagy) or delivering functional RNase T2 to the CNS is achievable, addressing the upstream storage lesion rather than only the inflammatory consequence.
7. **Human tissue validation.** Where autopsy/biopsy material is available, confirm microglial pyroptosis, ISG expression, and CD8+ T-cell infiltration in patient brain to close the model-to-human gap.

Report compiled from 14 confirmed findings across 5 iterations, drawing on 14 primary and review references. Evidence types are annotated per finding (human clinical/genetic, model organism, in vitro, review). Ontology suggestions: MONDO:0013058, HGNC:14015 (RNASET2), UniProt O00584, GO:0005764 (lysosome), GO:0060337 (type I IFN signaling), GO:0070269 (pyroptosis), UBERON:0000955/0002316/0001871/0002421, CL:0000129 (microglia)/0000540 (neuron)/0000128 (oligodendrocyte)/0000127 (astrocyte).
